

115.109

Introduction to

Business Analytics

Topic 10:

Test 2 Prep

Dr Andrew Murphy

(Please contact me via Communication Tools in Stream)

Week (starting)	Business Problem	Data & Analysis	Chapter (Black et al)	Assess ments
1 (13/7)	Understanding Data: discounts & margin	Calculating data in Excel; GST, discounts and Margins	1	Lab1
2 (20/7)	Operational Monitoring: Dashboards	Tabulating & Graphing	2	Lab2
3 (27/7)	Industry trends	Statistics NZ: retail survey Descriptive stats: averages	3	Lab3
4 (3/8)	Customer trends	Statistics NZ: census Descriptive stats: variation	3	Lab4
5 (10/8)	Transaction data: product sales	Transaction data sources. Sampling	7	Lab5
6 (17/8)	Loyalty cards	Loyalty card schemes. Probability; proportions, χ^2	4, 12	Lab6 Test 1
Mid	Semester	Study	Period	
7 (7/9)	Segmentation	Statistics NZ: census. Normal distribution; Means	6, 8	Lab7
8 (14/9)	Determining difference (1)	Means; ANOVA	9, 10, 11	Lab8
9 (21/9)	Associating and Predicting	Explaining relationships: Correlation & regression	13	Lab9
10 (28/9)	Determining Difference (2)	Explaining relationships (continued)	14	-
11 (5/10)	Trends over time	Time series, Indices Unstructured data: social media	15	Lab11 Test 2
12 (12/10)	Corporate reporting	Finance Dashboards reporting		Report

Assessments Schedule

2026	Assessment	Lab 1&2	Lab 3&4	Lab 5&6	Test 1	Lab 7&8	Lab 9,11	Test 2	Report
Week	13-Jul	O: 13.07							
commencing	20-Jul		O: 20.07						
Monday:	27-Jul								
	03-Aug	D: 03.08							
	10-Aug			O: 10.08					
	17-Aug		D: 17.08		O: 14.08 C: 21.08				O 17.08
	<i>Mid-Semester Study Period</i>			D:31.08					
	07-Sep					O: 07.09			
	14-Sep								
	21-Sep						O: 21.09		
	28-Sep								
	05-Oct					D: 05.10		O: 02.10 C: 09.10	
	12-Oct						D: 19.10		D: 19.10
		O=opens (available from); D=due (without late penalty); C=closes (not available after)							

Test 2: 10% of final mark

- Opens Friday 2th October at 5pm, available until Friday 9th at 5pm.
- 1 hour maximum from commencement until submission – cannot be paused or restarted.
- Can be started anytime until 4:59pm on the 9th (in which case you will have 1 minute to complete the test)
- 20 multiple choice (1 mark each, collectively worth 10 out of 10) from week 6 (cross-tabs / chi-square) to week 9 (regression)

Report: citing data & summary report



SustainabilityData survey data (July 2020)

As used in the labs.

Citation: Perceptive (2020a) Sustainability survey data from the July 2020 Omnibus Survey [data set]



Sustainability Survey summary report (Perceptive)

This summary report from Perceptive might give you an idea of what might be interesting to analyse.

Please note for these data Perceptive have applied a weighting variable to make the results representative of the NZ population. The results will therefore be different from what you would find if doing the same analysis with the (unweighted) lab data.

You can use these outputs in your report. Citation: Perceptive (2020b) Massey University Sustainability in Retail Omnibus Report, July. Data analysed and provided by Perceptive Ltd



Q. Of the ones you selected: Please rank them from most important to least important



Report: citing data & summary report



Sustainability Survey demographics crosstab analysis report (Perceptive)

Please note for these data Perceptive have applied a weighting variable to make the results representative of the NZ population. The results will therefore be different from what you would find if doing the same analysis with the (unweighted) lab data.

See each spreadsheet tab for summaries of question groups by gender, age, income, ethnicity, region, plus employment and household status (which are two variables we do not have in the raw dataset)

You can use these outputs in your report. Citation: Perceptive (2020c) Sustainability Survey demographics crosstab analysis report. Data analysed and provided by Perceptive Ltd

Back to TOC											
What does sustainability mean to you? (select all that apply) by Banner											
	What best describes									Which of the following best	
Column %	Employed full-time	Employed part-time	Business Owner / Self-employed	Housewife / husband	Retired	Student	Unemployed	Other (please specify)	NET	Single person living alone	Single person living with partner
Being environmentally responsible	73%	76%	69%	76%	73%	66%	68%	67%	72%	80%	
Using renewable resources	63%	68%	70%	70%	75%	68%	61%	79%	67%	66%	
Protecting ecosystems	52%	53%	50%	54%	54%	56%	51%	56%	53%	60%	
Cleaning our waterways and oceans	39%	53%	45%	56%	60%	47%	48%	57%	48%	60%	
Targeting zero waste	44%	51%	52%	55%	49%	38%	43%	50%	46%	42%	
Conserving water	42%	51%	44%	48%	56%	30%	49%	57%	46%	52%	
Carbon emission reduction	47%	48%	46%	54%	43%	46%	37%	42%	46%	45%	
Being socially responsible	41%	48%	46%	38%	47%	31%	32%	32%	42%	43%	
Having actionable plans for climate change	36%	45%	29%	45%	33%	28%	37%	49%	37%	40%	
Traceability (can trace where the product has come from farm to fork)	23%	32%	29%	38%	38%	14%	29%	35%	28%	32%	
Limiting population and economic growth	22%	20%	24%	30%	28%	22%	26%	23%	23%	28%	
Having social well-being focus rather than economic progress	19%	22%	24%	18%	28%	27%	13%	34%	22%	23%	
Good corporate citizenship	16%	20%	20%	17%	25%	14%	9%	11%	18%	19%	
Not sure	2%	2%	4%	6%	4%	2%	11%	10%	3%	3%	
Other - please specify	0%	3%	5%	1%	3%	11%	1%	7%	2%	2%	
NET	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	
Column Population	430	144	74	64	182	62	66	19	1042	146	
Base Population	1042	1042	1042	1042	1042	1042	1042	1042	1042	1042	
Total sample: Weight: base n = 1040; total n = 1042											
Now open up by banner											
Please state by Banner	Please state how much you agree with the following statements Willingness to pay more Those who Agree + Strongly Agree by Banner										
To what extent by Banner	To what extent do you agree with the following statements1 Bottle deposit scheme - Agree + strongly agree by Banner										
Please state by Banner(2)	Please state how much you agree with the following statements Perceived customer effectiveness agree + strongly agree by Banner										
IXNow pleas by Banner	IX:Now please state how much you agree with the following statements - Sustainability literacy those think it is important + extremely important by Banner										
XHow importa by Banner	X:How important is it that a retail business/brand implements or are involved in the following: This section will cover the topic of Environmental Domain - those think it is important + extremely important by Banner										
XThis sectio by Banner	X:This section will cover the topic of Environmental Domain - those think it is important + extremely important by Banner										
XHow importa by Banner(2)	X:How important is it that a retail business/brand implements or are involved in the following: This section will cover the topic of Shareholder Domain - those think it is important + extremely important by Banner										
XThis sectio by Banner(2)	X:This section will cover the topic of Shareholder Domain - those think it is important + extremely important by Banner										
XHow importa by Banner(3)	X:How important is it that a retail business/brand implements or are involved in the following: This section will cover the topic of Employee Domain - those think it is important + extremely important by Banner										
XThis sectio by Banner(3)	X:This section will cover the topic of Employee Domain - those think it is important + extremely important by Banner										
This section by Banner	This section will cover the topic of Customer Domain - those think it is important + extremely important by Banner										

Report & Reflection

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DOC

Report Feedback Form



How reports will be assessed. I strongly encourage you to complete your own assessment ahead of time, to compare to the marked version.



ASSIGNMENT

Written Group Report (one submission per group: submitter does so on behalf of all members)



1 of 178 Submitted, 1 Ungraded

Opened 17 August 2026

Due 19 October 2026, 11:00 PM NZDT

Cut-off 27 October 2026, 11:00 PM NZDT



ASSIGNMENT

Individual Reflection on Group Work



This submission box is for EACH group member to submit their OWN INDIVIDUAL 500-word (maximum) reflection on how the group functioned: how roles were assigned and how well they were performed; how decisions were made and how any conflicts were managed.

Please use the Group Evaluation Template (see attached file), evaluating each member of the group. Your maximum 500-word reflection follows this form, including any comments to justify lower marks given to group members.

You should only lower a person's percentage if their level of contribution and engagement was significantly less than the entire group's, and/or they were unresponsive to other group member's efforts to engage them throughout the duration of the task. You can make this contribution as low as necessary, to reflect their overall contribution, including down to 0% if warranted.

Low scores will be taken into consideration when allocating marks to each member for the group report mark.

NB: if you ended up in a group of one, because your fellow group members never responded or your group fell apart, you still need to complete an individual reflection. You should discuss the attempts you made to contact and involve your group mates, including screen shots of key messages sent/received

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Review 3 Q1

A crosstab of Age and Household Income was constructed for data collected in a survey on Showrooming and Webrooming, as follows:

Age \ Income	<\$50k	\$50-100k	\$100-150k	>\$150k	Total
1 (18-24)	14	14	2	8	38
2 (25-34)	8	14	8	10	40
3 (35-44)	2	8	8	2	20
4 (45-54)	2	6	2	12	22
Total	26	42	20	32	120

Which statistical test is being performed, and why?

Select one:

- ☐ a. Chi-square: to test association / relationship between variables or bias in sample
- ☐ b. Coefficient of variation: to see how much variation there is around the mean
- ☐ c. Chi-square: to test whether age is caused by income
- ☐ d. A paired samples t-test: to check if age and income go together

An analyst constructed the following 'expected counts' from the above data:

Age \ Income	<\$50k	\$50-100k	\$100-150k	>\$150k	Total
1 (18-24)	8.2	13.3	6.3	10.1	38
2 (25-34)	8.7	14.0	6.7	10.7	40
3 (35-44)	4.3	7.0	3.3	5.3	20
4 (45-54)	4.8	7.7	3.7	5.9	22
Total	26	42	20	32	120

After taking the difference between each observed and expected count, squaring those differences, dividing each squared difference by the relevant expected count, and finally summing those values, the result came to 27.0.

Review 3

b) Construct appropriate Null and Alternative Hypotheses for this test

Select one:

- ☐ a. Ho: Income causes Age: they are not independent Ha: income does not cause Age: they are independent
- ☐ b. We need more information to create hypotheses for this data
- ☐ c. Ho: There is no association between Age and Income: they are independent Ha: There is an association between Age and Income: they are not independent
- ☐ d. Ho: There is an association between Age and Income: they are not independent Ha: There is no association between Age and Income: they are independent

Review 3

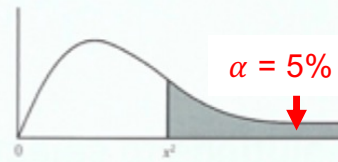
Age \ Income	<\$50k	\$50-100k	\$100-150k	>\$150k	Total
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4 (45-54)	2	6	2	12	22
Total	26	42	20	32	120

c) Determine the degrees of freedom associated with the rows and columns above

Select one:

- ☐ a. $(5-1)(5-1) = 16$ d.f.
- ☐ b. We need more information to determine the degrees of freedom
- ☐ c. $(4-1)(4-1) = 9$ d.f.
- ☐ d. $(4-1) + (4-1) = 6$ d.f.

d) Use the table below to determine the critical value, using an alpha value of 0.05



This table gives the critical values for various degrees of freedom ν and significance levels α

Table 9 Critical values for the chi-square distribution

α ν	0.10	0.05	0.025	0.01	0.005	0.001	α ν
1	2.706	3.841	5.024	6.635	7.879	10.827	1
2	4.605	5.991	7.378	9.210	10.597	13.815	2
3	6.151	7.815	9.348	11.345	12.838	16.266	3
4	7.779	9.488	11.143	13.277	14.860	18.466	4
5	9.236	11.070	12.832	15.086	16.750	20.515	5
6	10.645	12.592	14.449	16.812	18.548	22.457	6
7	12.017	14.067	16.013	18.475	20.278	24.321	7
8	13.362	15.507	17.535	20.090	21.955	26.124	8
9	14.684	16.919	19.023	21.666	23.589	27.877	9
10	15.987	18.307	20.483	23.209	25.188	29.588	10
11	17.275	19.675	21.920	24.725	26.757	31.264	11
12	18.549	21.026	23.337	26.217	28.300	32.909	12
13	19.812	22.362	24.736	27.688	29.819	34.527	13
14	21.064	23.685	26.119	29.141	31.319	36.124	14
15	22.307	24.996	27.488	30.578	32.801	37.698	15
16	23.542	26.296	28.845	32.000	34.267	39.252	16
17	24.769	27.587	30.191	33.409	35.718	40.791	17
18	25.989	28.869	31.526	34.805	37.156	42.312	18
19	27.204	30.144	32.852	36.191	38.582	43.819	19
20	28.412	31.410	34.170	37.566	39.997	45.314	20
21	29.615	32.671	35.479	38.932	41.401	46.796	21
22	30.813	33.924	36.781	40.289	42.796	48.268	22
23	32.007	35.172	38.076	41.638	44.181	49.728	23
24	33.196	36.415	39.364	42.980	45.558	51.179	24
25	34.382	37.652	40.646	44.314	46.928	52.619	25
26	35.563	38.885	41.923	45.642	48.290	54.051	26
27	36.741	40.113	43.195	46.963	49.645	55.475	27
28	37.916	41.337	44.461	48.278	50.994	56.892	28
29	39.087	42.557	45.722	49.588	52.335	58.301	29
30	40.256	43.773	46.979	50.892	53.672	59.702	30
40	51.805	55.758	59.342	63.691	66.766	73.403	40
50	63.167	67.505	71.420	76.154	79.490	86.660	50
60	74.397	79.082	83.298	88.379	91.952	99.608	60
70	85.527	90.531	95.023	100.425	104.215	112.317	70
80	96.578	101.979	106.629	112.329	116.321	124.839	80
90	107.565	113.145	118.136	124.116	128.299	137.208	90
100	118.498	124.342	129.561	135.807	140.170	149.449	100

Select one:

- ☐ a. 0.05
- ☐ b. 23.589
- ☐ c. 14.684
- ☐ d. 16.919

Review 3

After taking the difference between each observed and expected count, squaring those differences, dividing each squared difference by the relevant expected count, and finally summing those values, the result came to 27.0.

e) What can you conclude about the hypotheses in b) above?

Select one:

- ☐ a. Since $\chi^2 > \chi^2_c$, we cannot reject the null hypothesis: age and income are independent, and the sample is not biased.
- ☐ b. Since χ^2 exceeds 1.96, we reject the null hypothesis and accept the alternative: age and income are not independent, and the sample is biased.
- ☐ c. Since $\chi^2 > \chi^2_c$, we reject the null hypothesis and accept the alternative: age and income are not independent, and the sample is biased.
- ☐ d. Since χ^2 exceeds 1, the analyst must have made a mistake in the calculations, since this number should not be bigger than 1.

Review 3

Age \ Income	<\$50k	\$50-100k	\$100-150k	>\$150k	Total
1 (18-24)	8.2	13.3	6.3	10.1	38
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4 (45-54)	4.8	7.7	3.7	5.9	22
Total	26	42	20	32	120

f) If an assumption of this statistical test is that no more than 20% of cells should have an expected count less than 5.0, is this a problem for the above data? Suggest one way this could be resolved, if so.

Select one:

- ☐ a. Yes this is a problem as 4/16 (25%) have expected counts <5. We could group some categories together.
- ☐ b. Yes this is a problem as 5 of the observed counts are < 5. We could increase the sample size by 5 to fix.
- ☐ c. No this is not a problem as 4/25 (16%) have expected counts <5
- ☐ d. Yes this is a problem as 12 of the cells are > 5. We could recalculate the expected counts.

Review 4

According to the New Zealand Herald (article published 23 September 2018), New Zealanders request assistance from the New Zealand Government while visiting a range of countries at notably different rates (expressed per 100,000 NZ visitors), as per the following table.*

*(Unlike the original table, here the data are separated into groups of developed and developing countries, and some further calculations have been done for these two groups and for all countries.)

Country	Count	NZ Visitors	Rate per 100,000
Australia	117	9528348	1.2
UK	79	911522	8.7
Italy	70	124924	56.0
USA	64	1298049	4.9
Germany	45	122696	36.7
France	36	143146	25.1
UAE	32	45157	70.9
Japan	26	228444	11.4
Greece	25	23841	104.9
Singapore	25	170967	14.6
Developed	519	12597094	4.12
Thailand	150	352797	42.5
China	102	713315	14.3
New Celdonia	78	82437	94.6
Indonesia	67	260933	25.7
Fiji	50	1121467	4.5
Cook Islands	48	681464	7.0
Vietnam	45	141691	31.8
Phillippines	35	167330	20.9
Samoa	32	439536	7.3
India	31	400389	7.7
Developing	638	4361359	14.63
Overall	1157	16958453	6.82

Review 4 Q1

Treating the "NZ Visitors" as a proportional weight, the standard deviations for the developed countries and developing countries groups were calculated as 9.36 and 15.74 respectively. Using the mean, standard deviation and sample size generated the following test result:

	<i>Variable 1</i>	<i>Variable 2</i>
Mean	4.12	14.63
Variance	87.60	247.61
Observations	10	10
Pooled Variance	167.60	
Hypothesized Mean Difference	0	
df	18	
t Stat	-1.815042	
P(T<=t) one-tail	0.04310866	
t Critical one-tail	1.73406361	
P(T<=t) two-tail	0.08621731	
t Critical two-tail	2.10092204	

Country	Count	NZ Visitors	Rate per 100,000
Australia	117	9528348	1.2
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UAE	32	45157	70.9
Japan	26	228444	11.4
Greece	25	23841	104.9
Singapore	25	170967	14.6
Developed	519	12597094	4.12
Thailand	150	352797	42.5
China	102	713315	14.3
New Celdonia	78	82437	94.6
Indonesia	67	260933	25.7
Fiji	50	1121467	4.5
Cook Islands	48	681464	7.0
Vietnam	45	141691	31.8
Phillippines	35	167330	20.9
Samoa	32	439536	7.3
India	31	400389	7.7
Developing	638	4361359	14.63
Overall	1157	16958453	6.82

1) Explain which statistical test has been conducted

Select one:

- ☐ a. chi-square test
- ☐ b. two-sample t-test
- ☐ c. ANOVA
- ☐ d. One-sample t-test

Review 4

2) Which null and alternative hypotheses are most appropriate for this test?

Select one:

- ☐ a. $H_0: \bar{x}_1 = \bar{x}_2$; [null hypothesis]; $H_A: \bar{x}_1 \neq \bar{x}_2$ (two-tailed) or $H_A: \bar{x}_2 > \bar{x}_1$ (one-tailed) [alternative hypothesis]
- ☐ b. $H_0: \bar{x}_1 \neq \bar{x}_2$; [null hypothesis]; $H_A: \bar{x}_1 = \bar{x}_2$ (two-tailed) [alternative hypothesis]
- ☐ c. Null hypothesis: the test is significant; Alternative hypothesis: the test is not significant
- ☐ d. Null hypothesis: the test is not significant; Alternative hypothesis: the test is significant

3) Explain why you chose a one- or two-tailed test [in your hypotheses]

Select one:

- ☐ a. I chose a two-tailed test because there is no theoretical or stated reason to expect one mean to be larger than the other
- ☐ b. I chose a two-tailed test because there are two means
- ☐ c. I chose a two-tailed test because the data compare developing countries and developed countries
- ☐ d. I chose a one-tailed test because it makes sense that developing countries might have more requests since they likely have greater cultural and economic differences to NZ

Review 4

	<i>Variable 1</i>	<i>Variable 2</i>
Mean	4.12	14.63
Variance	87.60	247.61
Observations	10	10
Pooled Variance	167.60	
Hypothesized Mean Difference	0	
df	18	
t Stat	-1.815042	
P(T<=t) one-tail	0.04310866	
t Critical one-tail	1.73406361	
P(T<=t) two-tail	0.08621731	
t Critical two-tail	2.10092204	

4) Using the results table, identify the two ways you can make a decision about your hypotheses

Select one:

- ☐ a. I should compare 1) the "t Critical one-tail" against the "t-Stat"; and 2) "P(T<=t) one-tail" against 1.96
- ☐ b. Depending on how my alternative hypothesis is written, I should compare 1) the "t Stat" against the "t Critical one-tail" [or two-tail]; and 2) the "P(T<=t) one-tail" [or two-tail] against alpha (0.05)
- ☐ c. I should compare 1) the Z-statistic against the t-statistic; and 2) the critical value of Z against the critical value of t
- ☐ d. I should compare 1) the "t Critical one-tail" against the "t Critical two-tail"; and 2) the "P(T<=t) one-tail" against "P(T<=t) two-tail"

Review 4

	Variable 1	Variable 2
Mean	4.12	14.63
Variance	87.60	247.61
Observations	10	10
Pooled Variance	167.60	
Hypothesized Mean Difference	0	
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t Critical two-tail	2.10092204	

5) Identify the key pieces of information in the results table that you require to do make your decision

Select one:

- ☐ a. I compare 1) the "t Stat" of -1.815042 against the "t Critical two-tail" of -2.10092204, and conclude the t-statistic is not more extreme, or 2) the "P(T<=t) two-tail" of 0.08621731 against $\alpha=0.05$, and conclude the P-value is not smaller. Either way, I conclude I cannot reject the null hypothesis (ie, I cannot say there is a significant difference in means).
- ☐ b. I compare 1) "t Critical one-tail" of -1.73406361 against the "t Critical two-tail" of -2.10092204, and conclude the one-tail is better, or 2) the "P(T<=t) one-tail" of 0.04310866 against the "P(T<=t) two-tail" of 0.08621731, and conclude the two-tail is better.
- ☐ c. I compare 1) the z-statistic of 0.05 against the t-statistic of -1.815042; and 2) the critical value of z of -1.96 against the critical value of t of -2.10092204, and conclude that the t-test is better
- ☐ d. I compare 1) "t Critical one-tail" of -1.73406361 against the "t-Stat" of -1.815042, and conclude the critical value is smaller; or 2) the "P(T<=t) one-tail" of 0.04310866 against 1.96, and conclude the P-value is smaller. Either way, I reject the null hypothesis and accept the alternative: the means are different.

Review 4

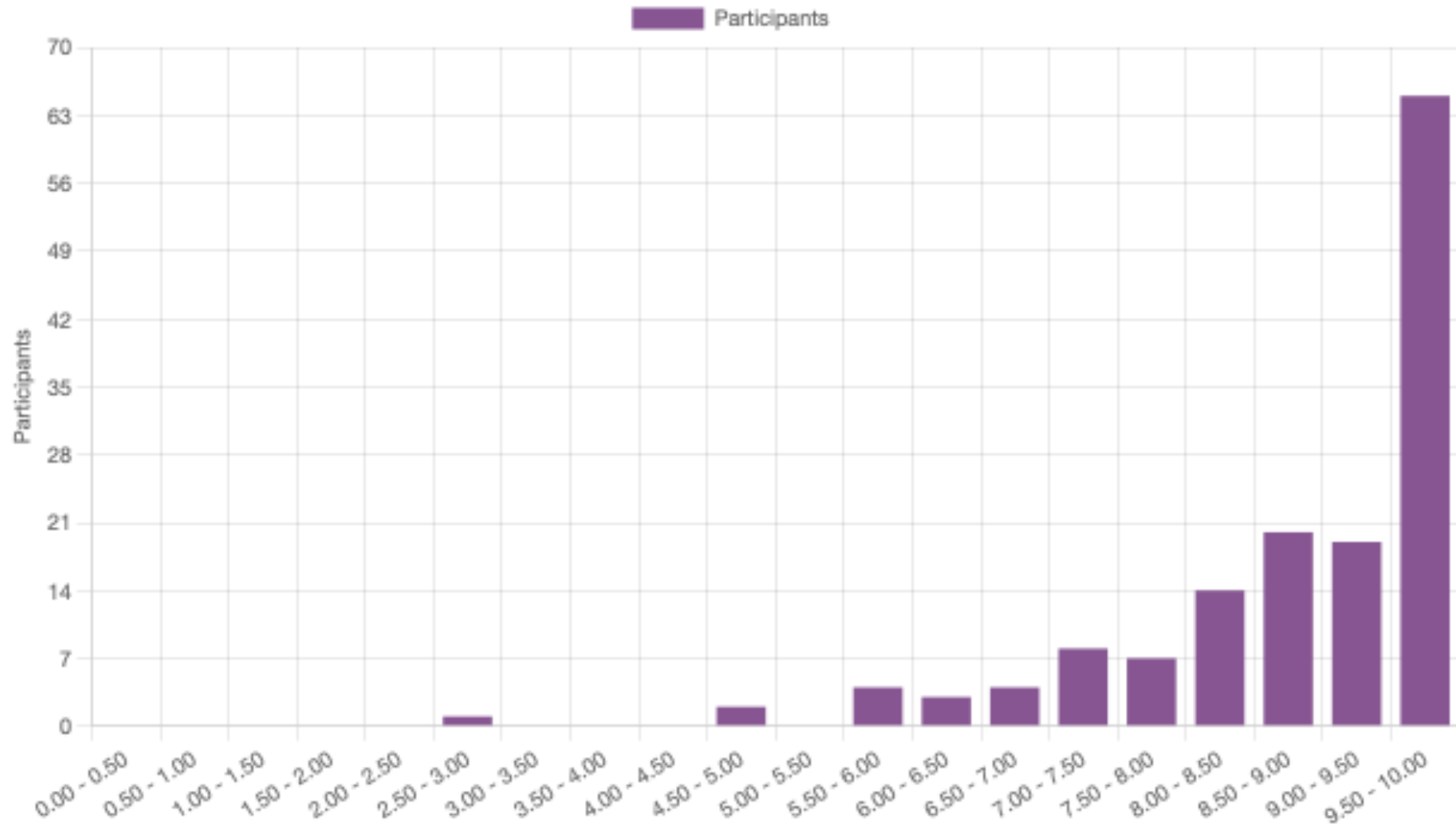
Country	Count	NZ Visitors	Rate per 100,000
Australia	117	9528348	1.2
UK	79	911522	8.7
Italy	70	124924	56.0
USA	64	1298049	4.9
Germany	45	122696	36.7
France	36	143146	25.1
UAE	32	45157	70.9
Japan	26	228444	11.4
Greece	25	23841	104.9
Singapore	25	170967	14.6
Developed	519	12597094	4.12
Thailand	150	352797	42.5
China	102	713315	14.3
New Celdonia	78	82437	94.6
Indonesia	67	260933	25.7
Fiji	50	1121467	4.5
Cook Islands	48	681464	7.0
Vietnam	45	141691	31.8
Phillippines	35	167330	20.9
Samoa	32	439536	7.3
India	31	400389	7.7
Developing	638	4361359	14.63
Overall	1157	16958453	6.82

5) What can you therefore recommend to a travel insurance company for New Zealanders travelling to the group of developed countries and developing countries according to the data above?

Select one:

- ☐ a. It costs more to stay in developed countries, so products should be priced accordingly.
- ☐ b. Visitors should avoid Greece.
- ☐ c. Once weighted for visitor numbers, requests for Government assistance by NZers visiting developing countries are significantly greater (per visitor) than developed country requests (whether this reflects risk is another matter we cannot answer from the data). Travel insurance should price travel to developing countries more than developed if requests for government assistance is also reflected in travel insurance claims. However, since visitor total to the listed developing countries is less than a quarter that for the listed developed countries, the overall risk for government and insurance companies is much less.
- ☐ d. More visitors go to developed countries, so the risk for government and insurance companies is much less for developing countries.

Test 2 Prep: Test 1 results



$\bar{x}$ of those attempting: 8.82 (2025: 8.26; 2024: 8.13; 2023: 7.64)

Previously: ♂ $\bar{x}$: 8.3; ♀ $\bar{x}$: 8.2; Albany Internal $\bar{x}$: 8.5, Distance $\bar{x}$: 8.0

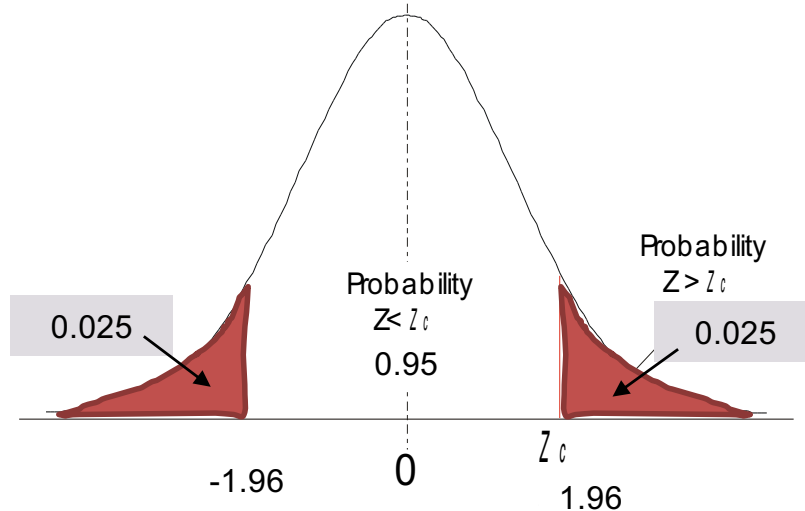
Test 2 Prep: Test 1 results

$\bar{x}$ of those attempting: 8.26 (2025)

♂ $\bar{x}$: 8.3; ♀ $\bar{x}$: 8.2; Albany Internal $\bar{x}$: 8.5, Distance $\bar{x}$: 8.0

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

	$\bar{x}$	n	s
All	8.26	126	1.466
Male	8.30	69	1.427
Female	8.21	58	1.510
AklInt	8.53	63	1.252
DL	7.98	63	1.617



Test 2 Prep Quiz: Q1



t-Test

 Anonymous

[Show Correct Answer](#)

[Show Responses](#)

A researcher conducts a two-tailed t-test to compare two samples, each of 100 observations, with the null hypothesis that the two sample means are the same and the alternative hypotheses that the first sample mean should be different from the second sample mean. If the calculated t-statistic is 1.980, the two-tailed critical value of t is 1.972, and the P value is .049, the researcher should:

A

reject the null hypothesis and reject the alternative hypothesis

B

not reject the null hypothesis

C

not reject the alternative hypothesis

D

reject the null hypothesis and accept the alternative hypothesis

E

increase the sample size

Test 2 Prep Quiz

Q2: The t-distribution has a mean of _____ and a standard deviation of _____

a) two; one

b) one; zero

c) one; two

d) zero; one

e) one; 1.96

Q3: The formula for the chi-square test uses:

a) observed and expected percentages

b) the two sample means

c) the standard error of the sample means

d) the two sample standard deviations

e) observed and expected frequency counts

Test 2 Prep Quiz

Q4: If there is no relationship between two variables, the correlation coefficient between them would be:

- a) unable to be calculated
- b) -1
- c) +1
- d) 0
- e) 0.5

Q5: If the calculated t-value is 1.98 and the critical value of t is 2.064, you should:

- a) increase the sample size
- b) reject the null hypothesis and reject the alternative hypothesis
- c) not reject the null hypothesis
- d) not reject the alternative hypothesis
- e) reject the null hypothesis and accept the alternative hypothesis

Test 2 Prep Quiz

Q6: If the regression equation is:
 $Y = 4.2 - 3X$, then the expected
score for Y when X is 4 would be:

- a) -16.2
- b) -7.8
- c) -12
- d) 7.8
- e) 13.8

Q7: The F-distribution is used in
which statistical test?

- a) Cross-tabulation
- b) ANOVA
- c) Confidence interval significance test
- d) Two-sample test of mean difference
- e) Test of difference in proportions

Test 2 Prep Quiz

Q8: If the regression equation is $Y = 24.35 - 14.2X$, then 24.35 is the _____, while -14.2 is the _____.

- a) y-intercept; slope
- b) x-intercept; y-intercept
- c) dependent variable; slope
- d) dependent variable; y-intercept
- e) slope; y-intercept

Q7 2016

7. The Net Promoter Score (NPS) has become a popular single indicator of customer loyalty toward a brand, product or company. A market research analyst has collected NPS data across brands and industries, and has summarised the distribution of a sample of recommendation scores for two industries, with recommendation rating scores as follows (Table 1), plus results for a test of means (generated using the Microsoft Excel 2013 Data Analysis ToolPak) (Table 2):

Table 1:

NPS Recommendation Ratings (0-10)		
Score ▼	Industry 1	Industry 2
0	2	1
1	6	1
2	2	0
3	5	1
4	3	4
5	17	8
6	14	5
7	27	28
8	34	31
9	34	20
10	56	30
Total	200	129

Table 2:

t-Test: Two-Sample Assuming Equal Variances		
	Variable 1	Variable 2
Mean	7.665	7.86046512
Variance	5.72138191	3.58975291
Observations	200	129
Pooled Variance	4.88698279	
Hypothesized Mean Difference	0	
df	327	
t Stat	-0.7829982	
P(T<=t) one-tail	0.21709743	
t Critical one-tail	1.64952678	
P(T<=t) two-tail	0.43419486	
t Critical two-tail	1.96724511	



Table 1:

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t Critical two-tail	1.96724511	

Next, collapse the recommend scale into three groups: 0-6 (labelled Detractors), 7-8 (Passives) and 9-10 (Promoters). [Q7 continued...]

The third step is to subtract the percentage of the sample who are Detractors from the percentage who are Promoters.

Topic 2
slides
57-60

- Calculate the Net Promoter Score for each industry, and interpret the results. [4 marks]
- Interpret the descriptive statistics shown in Table 2. [4 marks]
- Consider the t-test results shown in Table 2, and construct suitable null and alternate hypotheses that the analyst might have had in mind when conducting this test. [2 marks]
- Interpret the t-test results in light of your hypotheses. [3 marks]